Dense-Angle Two-Moons Benchmark

Architecture

This follow-up benchmark replaces the original 4-qubit HEA with an online-derived two-moons circuit. The adopted structure is the 2-qubit dense-angle classifier used in two-moons tutorials: four trainable input-scaling gates,

$$R_Y(w_0x_0),\ R_Y(w_1x_0),\ R_Z(w_2x_1),\ R_Z(w_3x_1),$$

followed by 4 repeated blocks of

$$R_Y(\theta_1)R_Y(\theta_2)\,\mathrm{CZ}\,R_Y(\theta_3)R_Y(\theta_4),$$

and a final pair of R_Y gates measured with Z_0. The data preprocessing also follows the centered angle map used in those examples, sending each feature to $[-\pi, \pi]$ on the training split.

Results

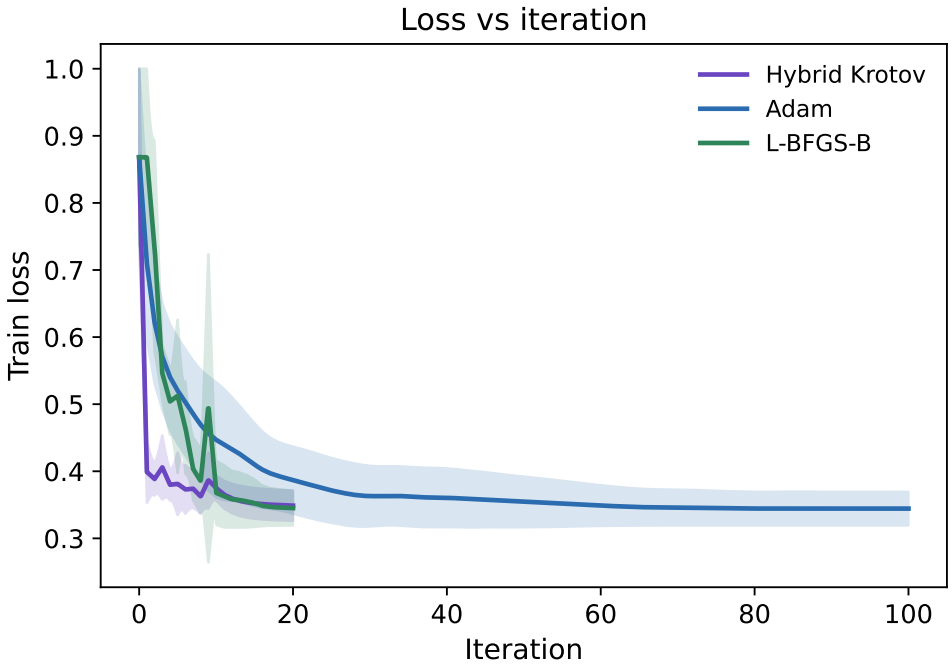
The optimizer comparison was rerun for Adam, L-BFGS-B, and the best previously discovered hybrid Krotov schedule, with the Krotov step sizes retuned for the new circuit. The aggregate results over 5 seeds are:

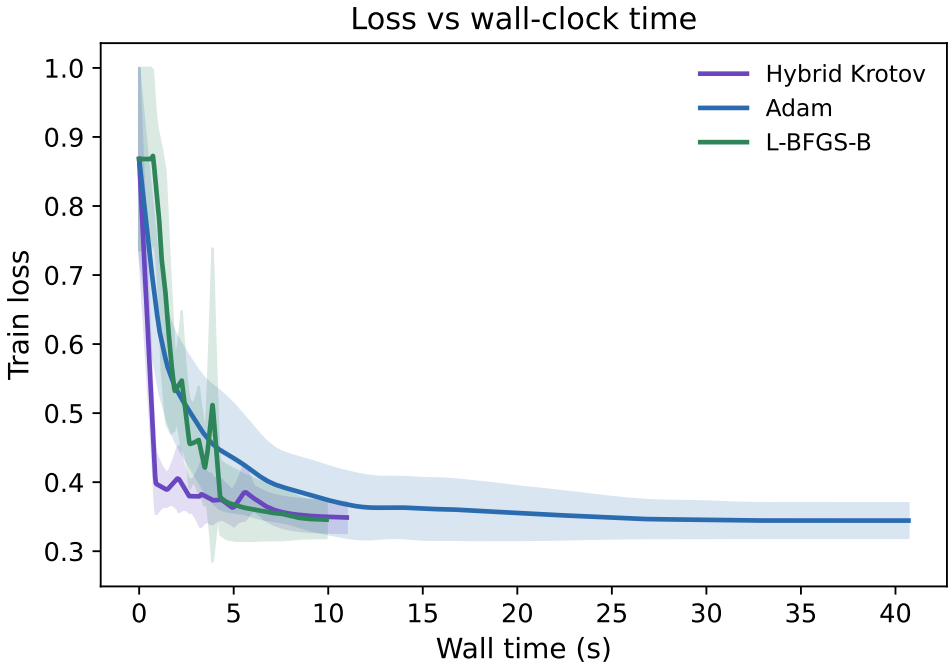
Optimizer	Final loss	Test accuracy	Wall time (s)	Time to $L \leq 0.40$ (s)
Hybrid Krotov	0.346 ± 0.020	0.871 ± 0.015	28.9 ± 13.0	1.22
Adam	0.344 ± 0.025	0.879 ± 0.020	40.8 ± 0.1	9.35
L-BFGS-B	0.344 ± 0.025	0.880 ± 0.019	13.3 ± 3.0	4.77

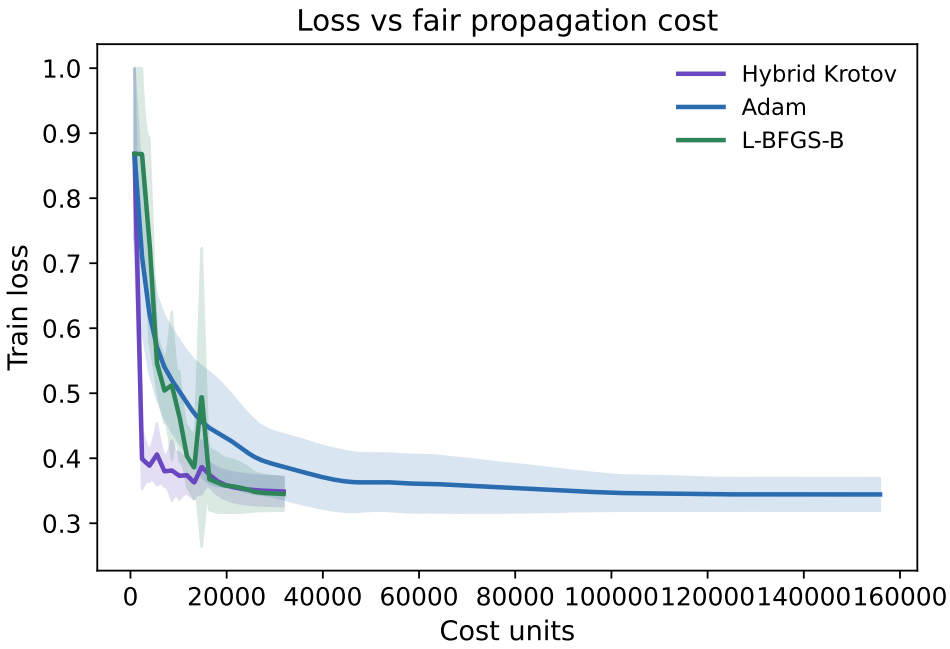
The best mean final loss in this architecture is achieved by **L-BFGS-B**, although Adam is statistically almost tied. In this corrected rerun, hybrid Krotov reaches mean final loss 0.346 and mean test accuracy 0.871, versus 0.344 / 0.879 for Adam and 0.344 / 0.880 for L-BFGS-B.

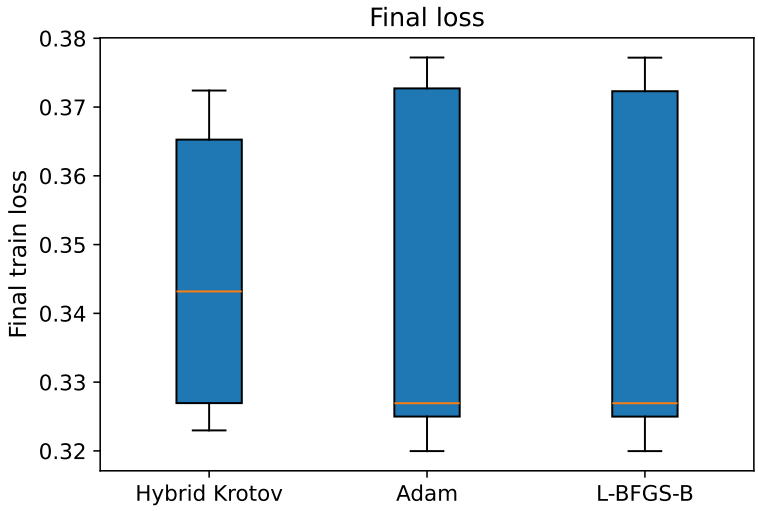
The strongest remaining advantage of hybrid Krotov is early progress: it is the fastest method to reach the benchmark threshold $L \leq 0.40$, while **L-BFGS-B** is the fastest optimizer in total wall-clock time. The corrected gradients therefore change the scientific conclusion substantially: the dense-angle architecture does help, but it no longer makes hybrid Krotov dominate the corrected classical baselines.

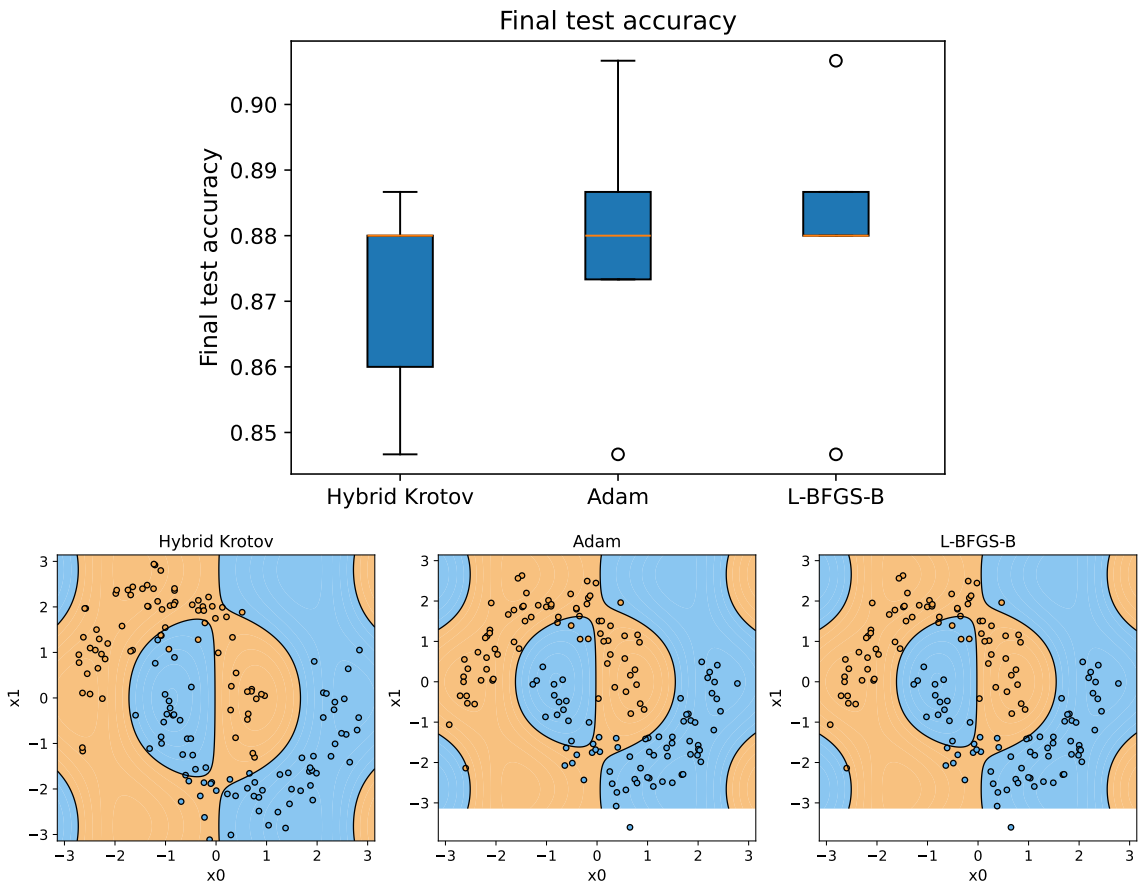
Figures


Loss vs iteration
Hybrid Krotov
Adam
L-BFGS-B
Train loss
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
20
40
60
80
100
Iteration


Loss vs wall-clock time
Hybrid Krotov
Adam
L-BFGS-B
Train loss
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
5
10
15
20
25
30
35
40
Wall time (s)


Loss vs fair propagation cost
Hybrid Krotov
Adam
L-BFGS-B
Train loss
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0
20000
40000
60000
80000
100000
120000
140000
160000
Cost units


Final loss
Final train loss
0.38
0.37
0.36
0.35
0.34
0.33
0.32
Hybrid Krotov
Adam
L-BFGS-B


Final test accuracy
Final test accuracy
0.90
0.89
0.88
0.87
0.86
0.85
Hybrid Krotov
Adam
L-BFGS-B
Hybrid Krotov
Adam
L-BFGS-B
x0
x1